Topic 1.12 - Transformations of Functions

Guided Notes

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: ______________________ Class: ____________ Date: ____________

Why this topic matters

Transformation notation compresses a great deal of geometric information. Strong reasoning distinguishes horizontal and vertical actions, explains the apparent reversal inside the input, maps points exactly, and tracks features such as domain, range, asymptotes, and excluded inputs rather than relying only on a memorized list of moves.

Learning targets

- Interpret and construct transformations in the form $g(x) = af(b(x-h)) + k$.
- Map points and key features between parent and transformed functions.
- Determine transformed domains, ranges, intercepts, asymptotes, and holes with justification.

Language and structure

Term	Meaning in this topic
horizontal translation	The shift controlled by h inside $f(b(x-h))$.
vertical translation	The shift controlled by k outside the function.
horizontal scale	Input distances are multiplied by $1/\lvert b\rvert$.
vertical scale	Output distances are multiplied by $\lvert a\rvert$.
point mapping	If (u, v) lies on f, then $(h + u/b,\ k + av)$ lies on g.

Launch - reason before calculating

A point $(-2, 5)$ lies on $y = f(x)$. Without graphing the full function, determine the corresponding point on $g(x) = -3f(2(x-1)) + 4$ and explain the role of every parameter.

Concept 1: Reading and mapping a general transformation

Core idea

For $g(x) = af(b(x-h)) + k$, solve $b(x-h) = u$ to map an original input u. Thus (u, v) maps to $(h + u/b, k + av)$. A negative b reflects horizontally, a negative a reflects vertically, and the horizontal scale is reciprocal.

Worked example - annotate the reasoning

Problem. If $(6, -1)$ lies on f, find its image on $h(x) = 2f((x+4)/3) - 5$.

Model reasoning. Set $(x+4)/3 = 6$, so $x = 14$. The output becomes $2(-1) - 5 = -7$. The image is $(14, -7)$.

Check your understanding

Independent | No calculator

If $(-3, 4)$ lies on f, find its image on $g(x) = -2f(-4(x-1)) + 7$.

Concept 2: Transforming parent graphs, domains, and ranges

Core idea

Transform key features, not just isolated points. Vertical shifts change range and horizontal asymptotes; horizontal shifts change domain landmarks and vertical asymptotes. Reflections and stretches change orientation and scale. For a quadratic in vertex form, the transformed vertex and axis are immediately visible.

Worked example - annotate the reasoning

Problem. Analyze $g(x) = -2(x+3)^2 + 5$.

Model reasoning. From $y = x^2$, shift left 3, stretch vertically by 2, reflect across the x-axis, and shift up 5. Vertex $(-3, 5)$, axis $x = -3$, domain all real numbers, range $y \leqslant 5$.

Check your understanding

Independent | No calculator

Analyze $q(x) = 3|x-2| - 4$, including vertex, opening, domain, and range.

Concept 3: Asymptotes, holes, and transformed restrictions

Core idea

For $r(x) = a/(x - h) + k$, the vertical asymptote is $x = h$ and horizontal asymptote is $y = k$. A transformed excluded input is found by solving the inside expression equal to the original excluded input. A transformed hole's coordinate follows the same point mapping as any other feature.

Worked example - annotate the reasoning

Problem. Analyze $r(x) = -\frac{2}{x-4} + 3$.

Model reasoning. Vertical asymptote $x = 4$, horizontal asymptote $y = 3$, domain $x \neq 4$, range $y \neq 3$. The negative scale reflects the reciprocal branches; for $x > 4$, values lie below 3.

Check your understanding

Independent | No calculator

For $s(x) = \frac{3}{2(x+1)} - 5$, identify asymptotes, domain, and range.

Representation laboratory

Connect at least two representations

The table contains points of f: $(-4, 2), (-2, -1), (0, 3), (6, 5)$. Construct the corresponding table for $g(x) = 2f((x - 1)/3) - 4$ and describe how input spacing and output spacing change.

Challenge laboratory

Challenge - justify every claim

A transformed function $g(x) = af(b(x-h)) + k$ maps the points $(0,2)$ and $(4,-1)$ on f to $(5,7)$ and $(3,1)$ on g. Determine a, b, h, k if $a > 0$ and explain whether the parameters are unique from the stated information.

Error clinic and calculator strategy

Common errors to diagnose

- Treating $f(2x)$ as a horizontal stretch by 2 instead of a compression by $1/2$.
- Shifting a graph before accounting for the full grouped input $b(x-h)$.
- Transforming points but forgetting to transform domain endpoints or excluded inputs.
- Moving a horizontal asymptote horizontally when a horizontal shift does not change its y-value.

Technology decision

Graphing technology is useful for checking a transformation, but a window can hide a vertex or asymptote. Predict mapped features first, then choose a window containing those features and verify that the graph agrees.

AP-style synthesis

Multi-part reasoning task

The points $(-4, 3), (-1, -2), (2, 5)$ lie on f. Define $g(x) = -2f(3(x-1)) + 4$.

A. Map all three points to g.

B. If the domain of f is $[-4, 2]$, determine the domain of g.

C. If the range of f is $[-2, 5]$, determine the range of g and explain the reversal of endpoints.

Exit ticket

Before you leave

1. What is the point map for $g(x) = af(b(x-h)) + k$?
2. Why does $f(4x)$ compress horizontally?
3. What are the asymptotes of $a/(x-h) + k$?

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